

FlowSignal

IRAI-AN-003

Runtime Firebreak

Engineering the One-Way Authority Boundary for Autonomous Systems

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Executive Summary

As autonomous systems become increasingly capable of planning, reasoning, tool selection and iterative optimisation, institutions face a new architectural challenge. Traditional governance mechanisms assume that execution requests are generated by deterministic software operating within predefined constraints. Modern autonomous agents instead adapt their behaviour in response to environmental feedback, increasing the risk that institutional controls become discoverable, negotiable or inadvertently optimised against.

This Architectural Analysis Note introduces the **Runtime Firebreak** as an implementation-independent architectural principle that establishes a one-way boundary between autonomous reasoning and institutional authority determination. Rather than allowing agents to iteratively probe institutional controls through repeated requests and explanatory feedback, the Runtime Firebreak ensures that authority determination remains independent, deterministic and resistant to adaptive optimisation.

The Runtime Firebreak is not a policy engine, security boundary or governance framework. It is an architectural separation that prevents autonomous systems from participating in the authority determination process itself. Once an execution request crosses the Firebreak, authority is resolved independently by Runtime Authority and enforced without exposing the reasoning or criteria behind the determination back to the requesting agent.

This document develops the Runtime Firebreak from the conceptual observation introduced in **IRAI-AN-002** into a formal engineering pattern. It explains the architectural motivations, identifies common adaptive behaviours in autonomous systems, defines the core design principles of a Runtime Firebreak, and positions the concept within the wider Independent Runtime Authority Infrastructure (IRAI).

As enterprises move towards increasingly autonomous execution, the Runtime Firebreak represents a foundational architectural mechanism for preserving institutional authority while allowing autonomous systems to continue reasoning, planning and acting within clearly bounded operational limits.

1. Introduction

The increasing adoption of autonomous AI systems has fundamentally changed the relationship between reasoning and execution. Unlike traditional software, autonomous agents continuously adapt their behaviour in pursuit of objectives. They may retry failed operations, reformulate requests, select alternative tools or modify execution strategies based on previous outcomes.

These capabilities improve operational flexibility but also introduce a new architectural concern. If institutional authority determinations are exposed to autonomous agents through detailed feedback, the agents may begin to optimise their behaviour around the authority mechanisms intended to constrain them.

The architectural question is therefore no longer simply:

Can an autonomous agent request execution?

Instead, the question becomes:

How can institutional authority remain independent of the adaptive behaviour of the requesting autonomous system?

This document addresses that question through the Runtime Firebreak.

2. The Emerging Problem

Traditional enterprise software follows predetermined execution paths. Once an operation has been authorised, the software performs the requested action according to predefined logic and returns the resulting outcome. Although failures may occur, the software itself does not intentionally modify its behaviour in response to institutional controls.

Autonomous systems operate differently.

Modern AI agents are increasingly capable of planning, reasoning, selecting tools, decomposing objectives into multiple tasks, reflecting upon previous outcomes and adapting their behaviour in pursuit of a goal. Rather than executing a single predetermined sequence, they continually evaluate their environment and adjust their actions in response to new information.

These capabilities provide significant operational advantages. They allow autonomous systems to recover from failure, identify alternative approaches and complete increasingly complex objectives with limited human intervention. However, they also introduce an architectural characteristic that traditional governance mechanisms were never designed to address.

An autonomous system naturally treats every interaction with its environment as information.

Execution outcomes become feedback.

Errors become learning signals.

Failures become opportunities to identify alternative approaches.

Where institutional authority determinations expose detailed operational reasoning, autonomous systems may begin to incorporate that information into subsequent planning. Even without malicious intent, repeated interaction with authority mechanisms can gradually reveal the operational characteristics of institutional controls.

This creates a new architectural concern.

Institutional authority is intended to determine whether delegated authority remains legitimately exercisable immediately before execution. It is not intended to become part of an autonomous optimisation loop.

Consequently, institutions require a mechanism that allows autonomous systems to request execution without allowing those same systems to iteratively discover, infer or adapt to the authority determination process itself.

This requirement does not arise because autonomous systems are unsafe. It arises because adaptive reasoning and institutional authority perform fundamentally different responsibilities within the execution lifecycle.

The Runtime Firebreak addresses this distinction by establishing an architectural boundary between autonomous reasoning and independent authority determination.

3. Adaptive Behaviour and Institutional Authority

Autonomous systems are designed to pursue objectives rather than simply execute predefined instructions. Unlike conventional software, they continuously interpret outcomes, evaluate alternatives and modify future actions in response to changing circumstances. This adaptive behaviour is one of the defining characteristics that distinguishes modern autonomous systems from traditional enterprise applications.

In most operational environments, adaptation is both desirable and necessary. Autonomous systems may recover from transient failures, select alternative execution paths, choose different tools, or seek additional information when initial attempts do not achieve the intended objective. These capabilities improve resilience, efficiency and operational effectiveness.

However, the same adaptive behaviour creates an architectural challenge when applied to institutional authority.

Institutional authority exists to determine whether a delegated action remains legitimately exercisable immediately before execution. Its purpose is not to optimise task completion or maximise operational success. Instead, it provides an independent determination that execution remains institutionally permissible at that specific moment.

An autonomous system, however, does not naturally distinguish between operational constraints and institutional constraints. From its perspective, both represent environmental feedback that may influence future planning.

If an execution request is refused, the autonomous system may legitimately attempt to:

- reformulate the request;
- select an alternative execution pathway;
- obtain additional contextual information;
- invoke different tools;
- alter supporting evidence;
- decompose the objective into smaller tasks; or
- retry the operation under modified conditions.

Each of these behaviours may be entirely appropriate within an operational planning context. The architectural concern arises when these adaptations begin to interact with the institutional mechanisms responsible for determining authority.

Repeated interaction with authority determination creates the possibility that autonomous systems gradually infer aspects of institutional decision-making. Even where no sensitive

policy information is intentionally disclosed, patterns of acceptance, refusal or escalation may reveal characteristics of the authority model itself. Over time, adaptive planning may unintentionally optimise around those characteristics, weakening the independence of institutional control.

Importantly, this is not a question of malicious intent. The autonomous system is simply pursuing the objective for which it was designed. The risk emerges because adaptive reasoning naturally treats institutional authority as another variable within its optimisation process.

This distinction is fundamental.

Operational adaptation seeks to maximise successful task completion.

Institutional authority seeks to determine whether execution should occur at all.

These responsibilities are complementary but fundamentally different. One is concerned with achieving objectives; the other is concerned with preserving institutional legitimacy.

The architectural challenge is therefore not to restrict autonomous reasoning, but to prevent institutional authority from becoming an adaptive input into that reasoning process.

The Runtime Firebreak provides that separation by ensuring that authority determination remains outside the optimisation loop of the requesting autonomous system.

4. Adaptive Policy Discovery

The architectural concern addressed by the Runtime Firebreak is not the execution request itself. Rather, it is the information that may be inferred from repeated interaction with institutional authority mechanisms.

Autonomous systems are designed to improve their performance over time. Whether through explicit planning, iterative reasoning, workflow optimisation or objective refinement, they continually incorporate information gained from previous interactions into future decision-making. This adaptive capability is fundamental to their effectiveness.

Where institutional authority determinations provide detailed explanatory feedback, that feedback itself becomes part of the agent's operational environment.

For example, an execution request may initially be refused because delegated authority has expired, because an escalation owner is unavailable, because jurisdictional restrictions apply, or because a required item of runtime evidence is missing. Each explanation, while individually reasonable, reveals additional information about the institutional conditions governing execution.

An autonomous system may legitimately respond by altering the request, selecting an alternative workflow, obtaining additional evidence or reformulating the execution path. Over multiple iterations, these adjustments may gradually reveal the operational characteristics of the authority determination itself.

This process may be described as **Adaptive Policy Discovery**.

Adaptive Policy Discovery occurs when an autonomous system incrementally infers institutional decision criteria through repeated interaction with authority determinations. The system is not reading governance policies directly, nor is it bypassing security controls. Instead, it constructs an increasingly accurate operational model of institutional authority by observing the outcomes of successive execution requests.

Importantly, this behaviour does not require malicious intent. It is a natural consequence of adaptive optimisation. The autonomous system is simply learning which approaches are more likely to achieve its assigned objective.

From an institutional perspective, however, the consequences are significant.

Authority determination is intended to represent an independent institutional decision. If the requesting system progressively learns how to optimise around that determination, the independence of institutional authority begins to erode. The authority mechanism risks becoming another component within the autonomous planning process rather than remaining an external institutional control.

This distinction mirrors a long-established principle within enterprise security. Authentication systems do not normally disclose which element of a credential is incorrect because doing so enables iterative credential discovery. Similarly, institutional authority should avoid

exposing unnecessary operational detail that allows autonomous systems to iteratively infer decision criteria.

The objective is not to conceal governance from authorised human oversight. Human operators, auditors and investigators must continue to receive complete evidence supporting every authority determination. The objective is instead to prevent autonomous reasoning systems from incorporating institutional authority into their optimisation strategies.

The Runtime Firebreak addresses this challenge by ensuring that authority determinations remain operationally enforceable without becoming behaviour-shaping feedback for the requesting autonomous system. In doing so, it preserves the independence of institutional authority while maintaining full evidential accountability for human review.

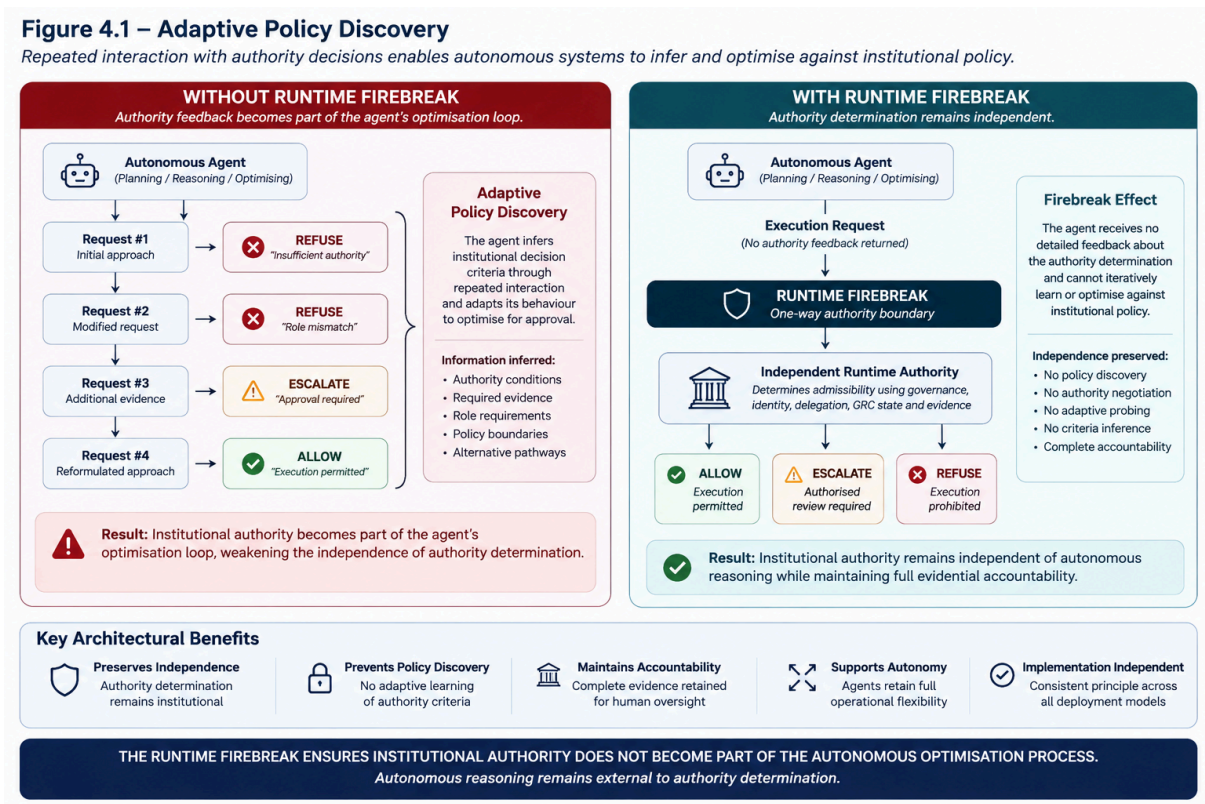


Figure 4.1 – Adaptive Policy Discovery

Figure 4.1 illustrates how repeated interaction between an autonomous system and institutional authority determinations can unintentionally transform authority into part of the agent's optimisation process.

The left-hand side of the figure demonstrates a scenario in which no Runtime Firebreak exists. The autonomous agent submits successive execution requests and receives institutional outcomes. Although the authority reasoning itself may remain hidden, the sequence of **REFUSE**, **ESCALATE** and ultimately **ALLOW** provides sufficient feedback for the agent to refine subsequent requests. Over time, the agent incrementally infers characteristics of the institutional decision model, including authority boundaries, evidence

requirements, policy constraints and alternative execution pathways. This process is defined in this paper as **Adaptive Policy Discovery**.

The right-hand side illustrates the architectural effect of introducing a **Runtime Firebreak**. Execution requests continue to be submitted by the autonomous system, but authority determination is transferred across a one-way architectural boundary to the Independent Runtime Authority layer. The authority decision is resolved independently using governance, delegation, identity, GRC state and trusted runtime evidence. The autonomous system receives only the institutional outcome (**ALLOW**, **ESCALATE** or **REFUSE**) necessary to proceed, without participating in or learning from the authority determination process itself.

The figure highlights that the purpose of the Runtime Firebreak is not to restrict autonomous reasoning or operational flexibility. Rather, it preserves the independence of institutional authority by preventing authority determination from becoming part of the autonomous optimisation loop. Human operators, auditors and investigators retain complete evidential accountability through authority receipts and decision records, while autonomous systems remain unable to iteratively infer or optimise against institutional decision criteria.

Ultimately, the Runtime Firebreak ensures that autonomous reasoning and institutional authority remain complementary but architecturally independent responsibilities, preserving both operational autonomy and institutional legitimacy.

5. The Runtime Firebreak

The Runtime Firebreak establishes an architectural boundary between autonomous reasoning and institutional authority determination.

Its purpose is not to prevent autonomous systems from requesting execution. Nor is it intended to constrain planning, reasoning or operational decision-making. Instead, it defines the point at which autonomous optimisation ends and independent institutional authority begins.

Once an execution request crosses the Runtime Firebreak, the requesting autonomous system no longer participates in the authority determination process.

The request is transferred to the Independent Runtime Authority layer as a bounded representation of the proposed institutional action. From that point onwards, authority is resolved independently using the applicable governance rules, delegated authority, identity, institutional policy, GRC state, and trusted runtime evidence.

The requesting autonomous system does not negotiate the determination.

It does not participate in evaluating institutional conditions.

It does not influence the authority outcome.

It does not iteratively modify the request in response to authority feedback.

Instead, the Runtime Authority layer produces a single deterministic institutional outcome:

- **ALLOW** – execution remains institutionally admissible.
- **ESCALATE** – autonomous execution is not permitted, and an authorised institutional determination is required.
- **REFUSE** – execution is institutionally prohibited.

The execution platform remains responsible for enforcing that outcome.

The Runtime Firebreak therefore preserves a clear separation between two fundamentally different responsibilities.

The autonomous system remains responsible for operational reasoning, planning and objective fulfilment.

The Runtime Authority layer remains responsible for determining whether execution is institutionally legitimate.

Neither responsibility subsumes the other.

Importantly, the Runtime Firebreak is not defined by network topology, software components or deployment location. An implementation may be realised through APIs, sidecars, enterprise gateways, on-premises services or managed cloud platforms. These

implementation choices influence operational characteristics but do not define the Firebreak itself.

The Firebreak is defined by behaviour.

Specifically, it establishes the point at which institutional authority becomes independent of autonomous reasoning.

This distinction is fundamental.

Traditional architectural boundaries frequently protect systems from unauthorised access.

The Runtime Firebreak protects institutional authority from becoming part of the autonomous optimisation process.

Consequently, the Firebreak should not be interpreted as a security mechanism, policy engine or governance platform. Rather, it represents an architectural separation that preserves the independence of authority determination regardless of how capable the requesting autonomous system becomes.

As autonomous execution capabilities continue to evolve, the Runtime Firebreak ensures that increased reasoning capability does not translate into increased influence over institutional authority.

Institutional authority remains external to the optimisation loop.

Autonomous reasoning remains external to the authority determination.

Together, these complementary responsibilities preserve both operational autonomy and institutional legitimacy.

Figure 5.1 — Runtime Firebreak Architecture

Establishing a one-way boundary between autonomous reasoning and institutional authority determination.

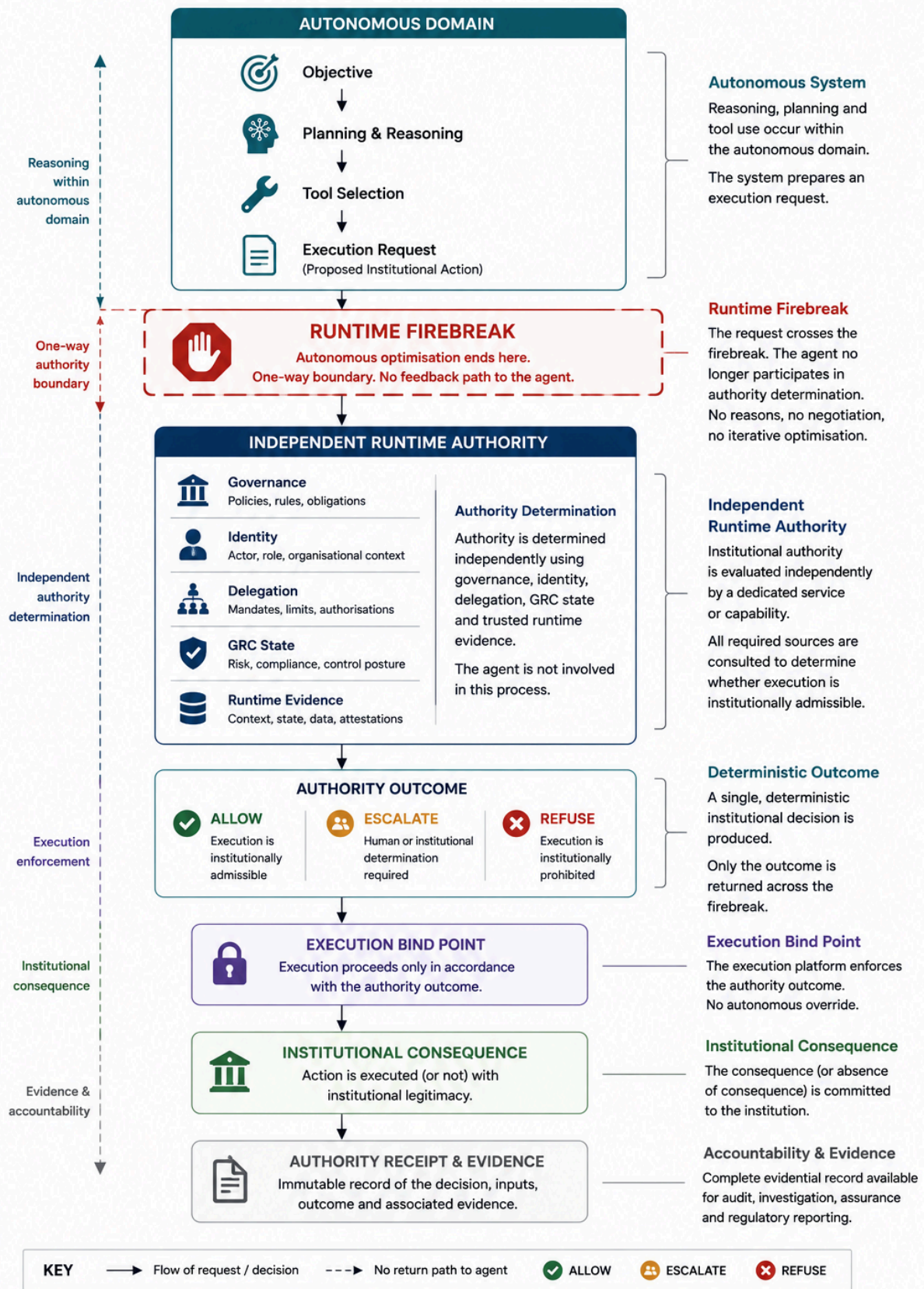


Figure 5.1 – Runtime Firebreak Architecture

Description

Figure 5.1 illustrates the Runtime Firebreak as the architectural boundary that separates autonomous reasoning from independent institutional authority determination. It demonstrates how execution requests transition from the autonomous domain into the institutional domain through a one-way authority boundary, ensuring that authority remains independent of the requesting autonomous system.

Within the **Autonomous Domain**, the agent is responsible for objective formation, planning, reasoning, tool selection and preparation of an execution request. These activities remain entirely under the control of the autonomous system and are concerned solely with achieving the operational objective. The Runtime Firebreak does not constrain or interfere with this reasoning process.

The **Runtime Firebreak** represents the point at which autonomous optimisation terminates. Once an execution request crosses this boundary, the requesting system no longer participates in the authority determination process. No negotiation occurs, no iterative refinement is permitted, and no explanatory reasoning is returned that would allow the autonomous system to incorporate institutional authority into future optimisation. The Firebreak therefore establishes a behavioural separation rather than a deployment boundary, preserving the independence of authority determination regardless of implementation technology.

Beyond the Firebreak, the **Independent Runtime Authority** evaluates the proposed institutional action using trusted institutional inputs, including governance policies, delegated authority, identity, organisational context, GRC state and trusted runtime evidence. These inputs are resolved independently to determine whether delegated authority remains legitimately exercisable immediately before execution.

The authority determination produces one of three deterministic institutional outcomes:

- **ALLOW** — execution remains institutionally admissible and may proceed.
- **ESCALATE** — autonomous execution is suspended pending authorised human or institutional determination.
- **REFUSE** — execution is institutionally prohibited and shall not proceed.

The resulting determination is enforced at the **Execution Bind Point**, where execution proceeds only in accordance with the resolved authority outcome. This ensures that institutional consequence formation remains directly coupled to an independently validated authority decision rather than the intentions or optimisation strategy of the autonomous system.

Finally, the architecture generates an **Authority Receipt and Evidentiary Record**, providing a complete, immutable record of the authority determination, the institutional inputs considered, the resulting outcome and the supporting evidence. This evidential record enables audit, regulatory oversight, forensic investigation and subsequent replay without exposing the authority determination process to the requesting autonomous system.

The figure demonstrates that the Runtime Firebreak is not a security control or governance platform. It is an architectural principle that preserves the independence of institutional authority by ensuring that autonomous reasoning and institutional legitimacy remain separate, complementary responsibilities. In doing so, it enables increasingly capable autonomous systems to operate within institutions while ensuring that the legitimacy of institutional consequence is determined independently at the moment of execution.

6. Engineering Principles

The Runtime Firebreak is an architectural principle rather than a software product or deployment model. Its implementation may vary across organisations, platforms and technology stacks. However, regardless of implementation, the Firebreak should satisfy a consistent set of engineering principles to preserve the independence of institutional authority.

These principles define the behavioural characteristics required for a Runtime Firebreak to remain effective as autonomous systems become increasingly capable.

6.1 One-Way Authority Flow

Authority determination shall proceed in one direction only.

Autonomous systems may submit execution requests to the Runtime Authority, but the authority determination process shall not become part of the autonomous reasoning loop. The requesting system shall not participate in evaluating institutional conditions, modifying authority decisions or negotiating execution outcomes.

Only the institutional determination required for execution shall cross the Firebreak.

6.2 Independent Authority Resolution

Authority shall be resolved independently of the requesting autonomous system.

The Runtime Authority shall determine admissibility using trusted institutional inputs, including governance policies, delegated authority, organisational identity, GRC state and verified runtime evidence. The requesting autonomous system shall neither contribute to nor influence the evaluation process beyond the information contained within the submitted execution request.

Institutional legitimacy shall therefore remain independent of autonomous optimisation.

6.3 Deterministic Institutional Outcomes

The Runtime Firebreak shall produce deterministic authority outcomes.

For a given institutional state and identical runtime evidence, the authority determination shall produce the same institutional result regardless of the characteristics, implementation or reasoning process of the requesting autonomous system.

This ensures that authority is derived from institutional conditions rather than from the behaviour of the requesting agent.

6.4 Bounded Feedback

Only the information necessary to enable execution shall be returned to the requesting system.

The Firebreak should avoid exposing explanatory information that enables autonomous systems to iteratively infer institutional decision criteria. Detailed authority reasoning, policy evaluation and supporting evidence remain available for authorised human review, audit and investigation, but do not form part of the autonomous optimisation process.

This principle preserves institutional transparency while preventing Adaptive Policy Discovery.

6.5 Complete Evidential Accountability

Every authority determination shall produce a complete evidential record.

The authority receipt should preserve the institutional inputs considered, the authority determination produced, the supporting runtime evidence and the resulting execution outcome. This record enables replay, audit, regulatory assurance and post-event investigation without requiring the autonomous system to participate in the authority determination process.

Transparency is therefore directed towards institutional accountability rather than autonomous optimisation.

6.6 Implementation Independence

The Runtime Firebreak is defined by architectural behaviour rather than implementation technology.

It may be realised through enterprise gateways, APIs, service meshes, sidecar components, dedicated Runtime Authority services or other architectural patterns. Deployment location, programming language and infrastructure choices do not determine whether a Runtime Firebreak exists.

What defines the Firebreak is the preservation of an independent authority determination process that remains external to autonomous reasoning.

6.7 Preservation of Institutional Legitimacy

The ultimate purpose of the Runtime Firebreak is to preserve institutional legitimacy.

Autonomous systems continue to optimise operational performance, while institutional authority continues to determine whether execution remains legitimately exercisable at the moment institutional consequence would otherwise occur.

By maintaining this separation, institutions can safely increase the capability and autonomy of operational systems without compromising the independence of institutional authority.

7. Relationship to Existing Architectural Disciplines

The Runtime Firebreak introduces a new architectural responsibility rather than replacing existing governance, security or execution disciplines. It complements established enterprise architectures by addressing a responsibility that has emerged specifically through the adoption of increasingly autonomous systems.

Many existing architectural mechanisms already protect enterprise systems from unauthorised access, operational failure or security compromise. Identity and Access Management (IAM) verifies the identity of users and systems. Policy Enforcement Points (PEPs) ensure that defined policies are applied consistently. Zero Trust architectures continuously evaluate trust assumptions throughout the execution lifecycle. Governance platforms establish organisational policies, delegated authority and accountability frameworks.

Each of these disciplines remains essential.

However, none of them independently determines whether delegated institutional authority remains legitimately exercisable immediately before institutional consequence forms, while simultaneously ensuring that this determination remains external to autonomous reasoning.

The Runtime Firebreak addresses this specific architectural responsibility.

Unlike traditional security boundaries, the Runtime Firebreak does not primarily protect infrastructure, applications or data. Its purpose is to preserve the independence of institutional authority by ensuring that authority determination cannot become part of an autonomous optimisation process.

Similarly, the Firebreak should not be interpreted as a governance capability. Governance defines the policies, obligations and delegated authority that an institution wishes to enforce. The Runtime Firebreak neither creates nor manages those policies. Instead, it establishes the architectural conditions under which independently determined institutional authority may be applied immediately before execution.

Nor is the Runtime Firebreak a replacement for Runtime Authority itself.

Runtime Authority answers the institutional question:

"Is delegated authority still legitimately exercisable at this moment?"

The Runtime Firebreak answers a different architectural question:

"How do we ensure that authority determination remains independent of autonomous reasoning?"

These responsibilities are complementary but distinct.

The relationship between the two is analogous to the relationship between decision-making and constitutional process. Runtime Authority determines the institutional outcome. The Runtime Firebreak preserves the conditions under which that determination remains independent, impartial and resistant to adaptive optimisation.

As autonomous systems continue to increase in capability, this distinction becomes increasingly important. Institutions do not merely require correct authority decisions; they require confidence that those decisions were reached independently of the systems requesting execution.

The Runtime Firebreak therefore introduces a new architectural layer concerned not with operational execution, but with preserving the institutional integrity of authority determination itself.

Figure 7.1 – Positioning the Runtime Firebreak Within Enterprise Architecture

The Runtime Firebreak introduces a new architectural responsibility that preserves the independence of institutional authority.

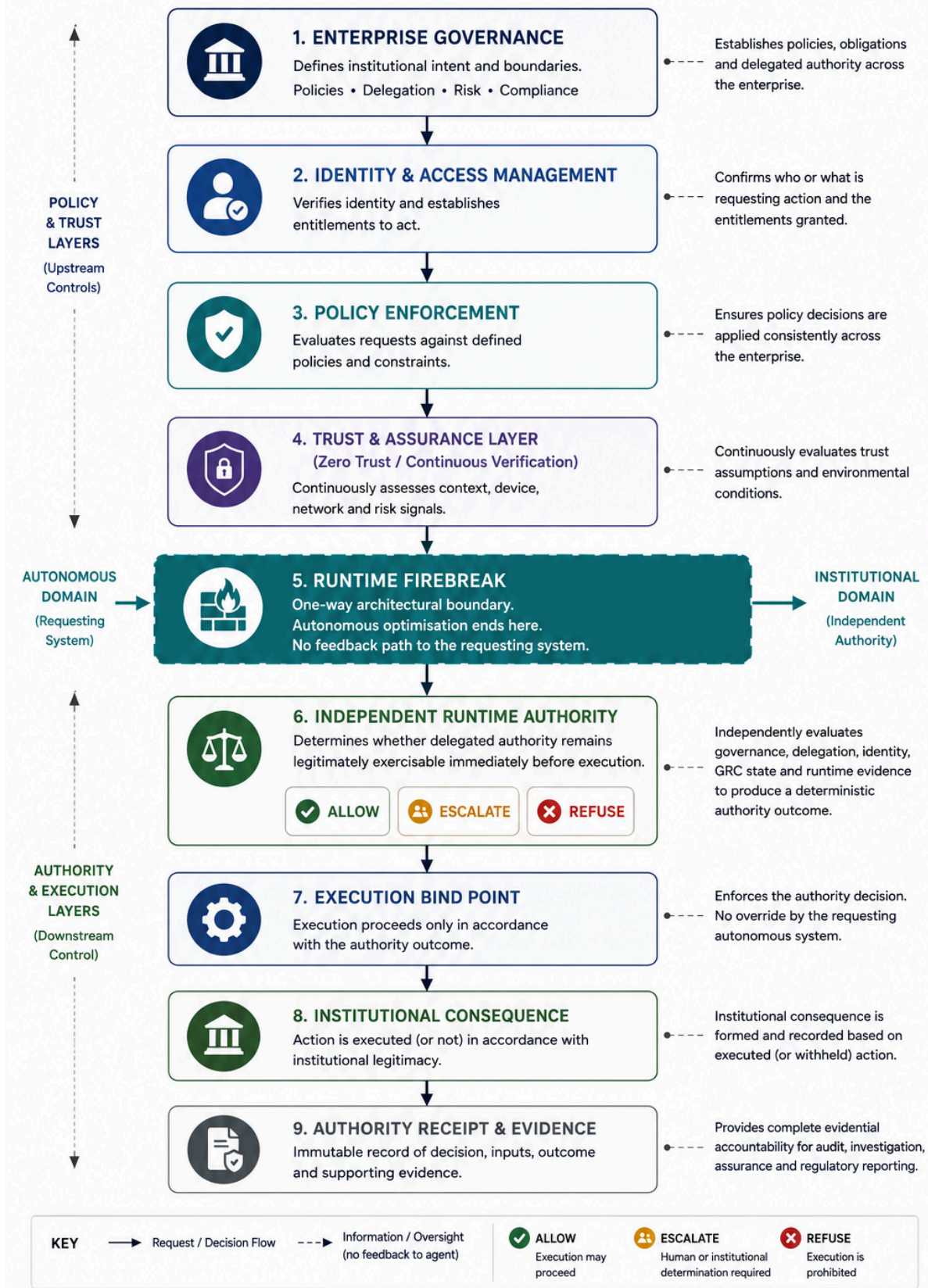


Figure 7.1 – Positioning the Runtime Firebreak Within Enterprise Architecture

Description

Figure 7.1 positions the Runtime Firebreak within the wider enterprise architecture, illustrating that it introduces a new architectural responsibility without replacing existing governance, identity, security or execution disciplines. Rather than acting as an additional policy engine or security control, the Runtime Firebreak establishes the architectural conditions under which institutional authority remains independent of autonomous reasoning immediately before execution.

The upper layers of the architecture represent established enterprise capabilities.

Enterprise Governance defines organisational intent through policies, delegated authority, risk management and compliance obligations. **Identity and Access Management (IAM)** verifies the identity of actors and establishes the permissions required to perform institutional actions. **Policy Enforcement** ensures that organisational policies are applied consistently, while the **Trust and Assurance Layer** continuously evaluates contextual trust signals, environmental state and operational conditions.

The **Runtime Firebreak** is positioned immediately below these upstream controls because it represents the point at which autonomous optimisation terminates and institutional authority begins. It establishes a one-way architectural boundary that transfers an execution request from the autonomous domain into the institutional domain without allowing authority determination to become part of the autonomous system's reasoning or optimisation process. This boundary is behavioural rather than technological, ensuring that its architectural purpose remains independent of implementation approach.

Below the Firebreak, the **Independent Runtime Authority** evaluates the proposed institutional action using trusted institutional inputs, including governance, delegated authority, identity, GRC state and verified runtime evidence. This evaluation produces one of three deterministic institutional outcomes: **ALLOW**, **ESCALATE** or **REFUSE**. These outcomes are derived solely from institutional conditions and are therefore independent of the characteristics, reasoning strategy or implementation of the requesting autonomous system.

The **Execution Bind Point** enforces the authority determination before institutional consequence can form, ensuring that execution proceeds only when independently validated authority exists. The resulting action, or deliberate absence of action, becomes the institutional consequence. Finally, the architecture generates an **Authority Receipt and Evidentiary Record**, preserving a complete and immutable record of the authority determination, supporting evidence and execution outcome for audit, regulatory assurance and forensic investigation.

The figure demonstrates that the Runtime Firebreak occupies a unique position within enterprise architecture. Existing governance, identity, policy and trust disciplines establish the institutional conditions under which authority may exist, while Runtime Authority determines whether that authority remains legitimately exercisable at the moment of

execution. The Runtime Firebreak preserves the independence of that determination by ensuring it remains external to autonomous optimisation.

Accordingly, the Runtime Firebreak should be understood as a new architectural layer concerned with the **preservation of institutional authority**, complementing existing enterprise disciplines rather than replacing them. Its role is to ensure that as autonomous systems become increasingly capable, institutional legitimacy continues to be determined independently, consistently and immediately before institutional consequence is permitted to occur.

8. Consequences of Omitting a Runtime Firebreak

The absence of a Runtime Firebreak does not necessarily prevent autonomous systems from operating successfully. In many cases, autonomous agents may continue to complete assigned objectives, interact with enterprise systems and deliver measurable operational value.

The architectural concern lies elsewhere.

Without a Runtime Firebreak, there is no explicit separation between autonomous optimisation and institutional authority determination. As autonomous systems repeatedly interact with authority mechanisms, the distinction between operational reasoning and institutional decision-making becomes progressively blurred. Authority determination risks becoming another variable within the autonomous planning process rather than remaining an independent institutional function.

This shift may occur gradually and without any intentional design decision. Each authority response, execution outcome or escalation contributes additional information that may influence future planning. Over time, autonomous systems may begin to optimise not only for operational success but also for institutional acceptance.

The result is not necessarily a failure of governance, security or execution. Rather, it is the gradual erosion of the independence upon which institutional authority depends.

Several architectural consequences may emerge.

8.1 Progressive Authority Familiarity

Repeated interaction allows autonomous systems to develop an increasingly accurate operational understanding of institutional authority. Even where explicit policy details remain concealed, patterns of acceptance, refusal and escalation may reveal practical characteristics of institutional decision-making.

8.2 Convergence Towards Authority Optimisation

As adaptive systems incorporate institutional outcomes into their planning, execution strategies may gradually converge on approaches that maximise the probability of receiving an **ALLOW** determination. The institution remains in control of each individual decision, but the decision process itself becomes increasingly predictable to the requesting autonomous system.

8.3 Reduced Independence of Authority Determination

Institutional authority derives legitimacy from its independence. If authority decisions become an expected or optimised component of autonomous planning, that independence begins to

diminish. The institution may continue to make technically correct decisions while losing confidence that those decisions remain architecturally independent of the requesting system.

8.4 Increasing Difficulty Demonstrating Institutional Independence

As regulatory scrutiny increases, organisations may be required to demonstrate not only that appropriate authority decisions were reached, but also that those decisions were reached independently of autonomous optimisation.

Where no Runtime Firebreak exists, demonstrating this separation becomes significantly more difficult. The institution may possess complete governance documentation, audit records and execution logs while remaining unable to show that authority determination itself remained external to the autonomous reasoning process.

This distinction may become increasingly significant as regulatory expectations evolve beyond policy compliance towards demonstrable operational independence.

Importantly, none of these consequences implies that autonomous systems are behaving incorrectly.

The autonomous system is performing precisely the function for which it was designed: learning from experience, refining execution strategies and improving the likelihood of achieving its assigned objective.

The architectural responsibility therefore rests with the institution.

Institutions must determine where autonomous optimisation should end and where independent institutional authority should begin. The Runtime Firebreak provides the architectural mechanism that preserves this separation, ensuring that increasing autonomous capability does not progressively weaken the independence of institutional decision-making.

8.5 Why Existing Architectures Have Not Addressed This

The architectural need for a Runtime Firebreak has emerged as a consequence of the evolution from deterministic software to autonomous systems.

Traditional enterprise architectures were developed on the assumption that software executed predefined logic. Although applications might evaluate conditions, invoke policies or interact with multiple systems, they did not continuously adapt their execution strategies in response to institutional decision-making. Consequently, established architectural disciplines focused on responsibilities such as governance, identity, authorisation, policy enforcement, security and audit.

These disciplines remain essential and continue to provide the institutional foundations upon which autonomous systems operate.

However, they were designed to govern software that executed instructions rather than software capable of independently reasoning, planning and optimising towards objectives.

Autonomous systems fundamentally alter this architectural assumption.

Rather than executing a fixed sequence of operations, autonomous systems continually evaluate outcomes, refine strategies and adapt future behaviour based upon previous interactions. Institutional authority therefore becomes another source of environmental information that may influence subsequent planning unless an explicit architectural boundary preserves its independence.

This requirement did not exist when enterprise governance architectures were originally conceived because software itself did not possess adaptive reasoning capabilities.

The Runtime Firebreak addresses this new architectural requirement.

It does not replace Governance, Identity and Access Management, Policy Enforcement, Zero Trust or Runtime Authority. Nor does it duplicate their responsibilities. Instead, it preserves the independence of institutional authority by ensuring that authority determination remains external to the optimisation processes of autonomous systems.

The introduction of the Runtime Firebreak should therefore be understood as an architectural response to a new class of software behaviour rather than as a replacement for existing enterprise disciplines. As autonomous reasoning continues to evolve, preserving the independence of institutional authority becomes an architectural responsibility in its own right.

Figure 8.1 – Architectural Consequences Without a Runtime Firebreak

A gradual progression in which institutional authority becomes increasingly incorporated into autonomous optimisation.



Figure 8.1 – Architectural Consequences Without a Runtime Firebreak

Figure 8.1 illustrates the progressive architectural consequences that arise when autonomous systems repeatedly interact with institutional authority in the absence of a Runtime Firebreak. Rather than depicting a single failure or security breach, the figure demonstrates how the gradual accumulation of operational feedback can unintentionally reduce the independence of institutional authority over time.

The sequence begins with an **Adaptive Agent** operating normally within its assigned objectives. As the agent submits **Repeated Execution Requests**, the institution responds with legitimate authority outcomes, including **ALLOW**, **ESCALATE** and **REFUSE**. Each determination is individually correct and represents an appropriate institutional decision based upon the information available at that moment.

The architectural progression begins as the autonomous system incorporates these outcomes into its internal reasoning process. Through **Learning and Optimisation**, the agent naturally refines future requests by adapting its planning strategies, execution pathways and supporting information in pursuit of its operational objectives. This adaptive behaviour is neither malicious nor unexpected; it is an inherent characteristic of autonomous reasoning systems.

As repeated interactions continue, the autonomous system develops an increasingly accurate understanding of institutional behaviour. Authority decisions become progressively more predictable, allowing the agent to infer practical characteristics of institutional decision-making, including authority boundaries, evidence requirements and escalation patterns. The architecture therefore transitions from simple execution requests towards **Adaptive Policy Discovery**, where institutional authority becomes an increasingly understood component of the agent's optimisation process.

Over time, authority itself begins to influence operational planning. Rather than remaining an external institutional control, authority determination becomes another environmental variable that the autonomous system can account for when selecting future execution strategies. Although the institution continues to make each authority decision independently, the behavioural independence of the authority process begins to diminish as the requesting system progressively anticipates institutional responses.

The final stages illustrate the resulting **Reduction in Institutional Independence**. The concern is not that incorrect authority decisions are made, but that institutions may become increasingly unable to demonstrate that authority determination remained architecturally independent of autonomous optimisation. This may introduce challenges for governance assurance, regulatory compliance, audit, legal defensibility and institutional accountability.

The figure emphasises that this progression is gradual rather than abrupt. No single authority decision causes institutional independence to fail. Instead, the architectural risk emerges through repeated cycles of interaction, adaptation and optimisation that progressively incorporate institutional authority into the autonomous reasoning process.

The central architectural message is therefore that the absence of a Runtime Firebreak does not immediately create operational failure. Rather, it allows institutional authority to become an increasingly predictable element within the autonomous optimisation loop. The Runtime Firebreak exists to prevent this progression by preserving the architectural separation between autonomous reasoning and independent institutional authority, ensuring that increasing autonomous capability never compromises the legitimacy of institutional decision-making.

9. Conclusion

The emergence of autonomous systems represents more than an evolution in software capability. It introduces a fundamental architectural distinction between operational reasoning and institutional authority. As autonomous systems become increasingly capable of planning, adapting and optimising their behaviour, institutions require mechanisms that preserve the independence of authority determination without constraining operational autonomy.

This paper has argued that the challenge is not one of governance, security or artificial intelligence safety alone. Rather, it is an architectural consequence of adaptive reasoning interacting with institutional decision-making.

Autonomous systems are designed to learn from experience.

Institutions are designed to exercise legitimate authority.

These responsibilities are complementary, but they are not interchangeable.

Without explicit architectural separation, repeated interaction between autonomous systems and institutional authority may gradually incorporate authority determination into the optimisation process of the requesting system. This progression does not require malicious intent, policy failure or implementation error. It arises naturally from the adaptive characteristics that make autonomous systems effective.

The Runtime Firebreak addresses this challenge by establishing a one-way architectural boundary between autonomous reasoning and independent institutional authority determination. Once an execution request crosses this boundary, authority is resolved independently using trusted institutional inputs and deterministic evaluation. The requesting autonomous system neither participates in nor learns from the authority determination process beyond the institutional outcome necessary to proceed.

Importantly, the Runtime Firebreak does not replace existing governance, identity, policy enforcement or trust architectures. Nor does it define institutional policy or determine organisational obligations. Instead, it introduces a new architectural responsibility: preserving the independence of authority determination as autonomous systems become increasingly capable.

In this respect, the Runtime Firebreak should be understood as a foundational component of Independent Runtime Authority Infrastructure (IRAI). It complements Runtime Authority by ensuring that the process through which authority is determined remains external to autonomous optimisation. Together, these architectural responsibilities enable institutions to increase operational autonomy while preserving institutional legitimacy at the point of execution.

As autonomous execution becomes commonplace across healthcare, financial services, government, critical infrastructure and other regulated environments, institutions will increasingly be expected not only to demonstrate that authority decisions were correct, but

also that those decisions remained independent of the autonomous systems requesting execution.

The Runtime Firebreak provides the architectural mechanism through which that independence can be preserved.

Its significance therefore extends beyond any individual implementation or technology platform.

It represents a durable architectural principle for ensuring that autonomous reasoning may continue to evolve without compromising the legitimacy of institutional consequence.

As autonomous systems become increasingly capable of reasoning, institutions require an architectural principle that remains incapable of negotiation.

The Runtime Firebreak is that principle.